

-- 1. Setup (Given Schema)

-- =====
-- BANKING DATABASE SETUP
-- =====

```
CREATE DATABASE banking_lab;
USE banking_lab;
-- Customers
CREATE TABLE customers (
  customer_id INT PRIMARY KEY AUTO_INCREMENT,
  customer_name VARCHAR(100),
  city VARCHAR(50)
);
-- Accounts
CREATE TABLE accounts (
  account_id INT PRIMARY KEY AUTO_INCREMENT,
  customer_id INT,
  account_type VARCHAR(50), -- Savings / Current
  balance DECIMAL(12,2),
  FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
);
-- Transactions
CREATE TABLE transactions (
  transaction_id INT PRIMARY KEY AUTO_INCREMENT,
  account_id INT,
  transaction_type VARCHAR(50), -- Deposit / Withdrawal
  amount DECIMAL(12,2),
  transaction_date DATE,
  FOREIGN KEY (account_id) REFERENCES accounts(account_id)
);
```

-- 2. Sample Data

```
-- Customers
INSERT INTO customers (customer_name, city) VALUES
('Aung Aung', 'Yangon'),
('Su Su', 'Mandalay'),
('Kyaw Kyaw', 'Yangon'),
('Mya Mya', 'Bago'),
('Ko Ko', 'Yangon');
-- Accounts
INSERT INTO accounts (customer_id, account_type, balance) VALUES
(1, 'Savings', 5000),
(2, 'Current', 8000),
(3, 'Savings', 12000),
(4, 'Savings', 3000),
(5, 'Current', 15000);
-- Transactions
INSERT INTO transactions (account_id, transaction_type, amount, transaction_date)
VALUES
```

```
(1, 'Deposit', 2000, '2025-01-01'),
(1, 'Withdrawal', 500, '2025-01-02'),
(2, 'Deposit', 3000, '2025-01-03'),
(3, 'Deposit', 4000, '2025-01-04'),
(3, 'Withdrawal', 1000, '2025-01-05'),
(4, 'Deposit', 1500, '2025-01-06'),
(5, 'Deposit', 7000, '2025-01-07'),
(5, 'Withdrawal', 2000, '2025-01-08');
```

62 -- 1. Display all customers
63 • SELECT * FROM customers;
64

customer_id	customer_name	city
1	Aung Aung	Yangon
2	Su Su	Mandalay
3	Kyaw Kyaw	Yangon
4	Mya Mya	Bago
5	Ko Ko	Yangon

65 -- 2. Show customer name and city only
66 • SELECT customer_name, city
67 FROM customers;
68

customer_name	city
Aung Aung	Yangon
Su Su	Mandalay
Kyaw Kyaw	Yangon
Mya Mya	Bago
Ko Ko	Yangon

69 -- 3. Display all accounts with balance
70 • SELECT account_id, customer_id, account_type, balance
71 FROM accounts;
72

account_id	customer_id	account_type	balance
1	1	Savings	5000.00
2	2	Current	8000.00
3	3	Savings	12000.00
4	4	Savings	3000.00
5	5	Current	15000.00

73 -- PART B: WHERE
74 -- 4. Find customers from Yangon
75 • SELECT *
76 FROM customers
77 WHERE city = 'Yangon';
78

customer_id	customer_name	city
1	Aung Aung	Yangon
3	Kyaw Kyaw	Yangon
5	Ko Ko	Yangon

79 -- 5. Find accounts with balance greater than 10,000
80 • SELECT *
81 FROM accounts
82 WHERE balance > 10000;
83

84 5. Show all Deposit transactions only

account_id	customer_id	account_type	balance
3	3	Savings	12000.00
5	5	Current	15000.00

84 -- 6. Show all Deposit transactions only

85 • SELECT *

86 FROM transactions

87 WHERE transaction_type = 'Deposit';

88

transaction_id	account_id	transaction_type	amount	transaction_date
1	1	Deposit	2000.00	2025-01-01
3	2	Deposit	3000.00	2025-01-03
4	3	Deposit	4000.00	2025-01-04
6	4	Deposit	1500.00	2025-01-06
7	5	Deposit	7000.00	2025-01-07
NULL	NULL	NULL	NULL	NULL

89 -- 7. Find transactions between '2025-01-01' and '2025-01-05'

90 • SELECT *

91 FROM transactions

92 WHERE transaction_date BETWEEN '2025-01-01' AND '2025-01-05';

93

transaction_id	account_id	transaction_type	amount	transaction_date
1	1	Deposit	2000.00	2025-01-01
2	1	Withdrawal	500.00	2025-01-02
3	2	Deposit	3000.00	2025-01-03
4	3	Deposit	4000.00	2025-01-04
5	3	Withdrawal	1000.00	2025-01-05
NULL	NULL	NULL	NULL	NULL

94 -- PART C: ORDER BY

95 -- 8. Display accounts sorted by balance (highest first)

96 • SELECT *

97 FROM accounts

98 ORDER BY balance DESC;

99

account_id	customer_id	account_type	balance
5	5	Current	15000.00
3	3	Savings	12000.00
2	2	Current	8000.00
1	1	Savings	5000.00
4	4	Savings	3000.00
NULL	NULL	NULL	NULL

100 -- 9. Show transactions sorted by date (latest first)

101 • SELECT *

102 FROM transactions

103 ORDER BY transaction_date DESC;

104

transaction_id	account_id	transaction_type	amount	transaction_date
8	5	Withdrawal	2000.00	2025-01-08
7	5	Deposit	7000.00	2025-01-07
6	4	Deposit	1500.00	2025-01-06
5	3	Withdrawal	1000.00	2025-01-05
4	3	Deposit	4000.00	2025-01-04
3	2	Deposit	3000.00	2025-01-03
2	1	Withdrawal	500.00	2025-01-02
1	1	Deposit	2000.00	2025-01-01
NULL	NULL	NULL	NULL	NULL

105 -- PART D: GROUP BY

106 -- 10. Total balance per account type

107 • SELECT account_type,

108 SUM(balance) AS total_balance

109 FROM accounts

110 GROUP BY account_type;

111

account_type	total_balance
Savings	20000.00
Current	23000.00

112 -- 11. Total transaction amount per account

113 • SELECT account_id,

114 SUM(amount) AS total_transaction_amount

115 FROM transactions

116 GROUP BY account_id;

117

account_id	total_transaction_amount
1	2500.00
2	3000.00
3	5000.00
4	1500.00
5	9000.00

118 -- 12. Number of transactions per account
 119 • SELECT account_id,
 120 COUNT(transaction_id) AS transaction_count
 121 FROM transactions
 122 GROUP BY account_id;
 123

account_id	transaction_count
1	2
2	1
3	2
4	1
5	2

126 -- 13. Show account types with total balance > 10,000
 127 • SELECT account_type,
 128 SUM(balance) AS total_balance
 129 FROM accounts
 130 GROUP BY account_type
 131 HAVING SUM(balance) > 10000;
 132

account_type	total_balance
Savings	20000.00
Current	23000.00

133 -- 14. Show accounts with total transaction amount > 5,000
 134 • SELECT account_id,
 135 SUM(amount) AS total_transaction_amount
 136 FROM transactions
 137 GROUP BY account_id
 138 HAVING SUM(amount) > 5000;
 139

account_id	total_transaction_amount
5	9000.00

141 -- 15. For each customer:
 142 -- • Show total transactions
 143 -- • Show total amount
 144 -- • Only include customers with total amount > 3,000
 145 -- • Sort by total amount descending
 146 • SELECT
 147 c.customer_id,
 148 c.customer_name,
 149 COUNT(t.transaction_id) AS total_transactions,
 150 SUM(t.amount) AS total_amount
 151 FROM customers c
 152 JOIN accounts a ON c.customer_id = a.customer_id
 153 JOIN transactions t ON a.account_id = t.account_id
 154 GROUP BY c.customer_id, c.customer_name
 155 HAVING SUM(t.amount) > 3000
 156 ORDER BY total_amount DESC;

customer_id	customer_name	total_transactions	total_amount
5	Ko Ko	2	9000.00
3	Kyaw Kyaw	2	5000.00

158 -- PART G: ADVANCED (JOIN + CLAUSES)
 159 -- 16. Show customer name, account type, and balance
 160 • SELECT
 161 c.customer_name,
 162 a.account_type,
 163 a.balance
 164 FROM customers c
 165 JOIN accounts a ON c.customer_id = a.customer_id;
 166

customer_name	account_type	balance
Aung Aung	Savings	5000.00
Su Su	Current	8000.00
Kyaw Kyaw	Savings	12000.00
Mya Mya	Savings	3000.00
Ko Ko	Current	15000.00

```
167 -- 17. Show total deposits per customer
168 • SELECT
169     c.customer_name,
170     SUM(t.amount) AS total_deposits
171 FROM customers c
172 JOIN accounts a ON c.customer_id = a.customer_id
173 JOIN transactions t ON a.account_id = t.account_id
174 WHERE t.transaction_type = 'Deposit'
175 GROUP BY c.customer_id, c.customer_name;
```

customer_name	total_deposits
Aung Aung	2000.00
Su Su	3000.00
Kyaw Kyaw	4000.00
Mya Mya	1500.00
Ko Ko	7000.00

```
177 -- 18. Show customers who made more than 1 transaction
178 • SELECT
179     c.customer_name,
180     COUNT(t.transaction_id) AS transaction_count
181 FROM customers c
182 JOIN accounts a ON c.customer_id = a.customer_id
183 JOIN transactions t ON a.account_id = t.account_id
184 GROUP BY c.customer_id, c.customer_name
185 HAVING COUNT(t.transaction_id) > 1;
```

customer_name	transaction_count
Aung Aung	2
Kyaw Kyaw	2
Ko Ko	2

```
187 -- 19. Show customers with total withdrawals > 1,000
188 • SELECT
189     c.customer_name,
190     SUM(t.amount) AS total_withdrawals
191 FROM customers c
192 JOIN accounts a ON c.customer_id = a.customer_id
193 JOIN transactions t ON a.account_id = t.account_id
194 WHERE t.transaction_type = 'Withdrawal'
195 GROUP BY c.customer_id, c.customer_name
196 HAVING SUM(t.amount) > 1000;
```

customer_name	total_withdrawals
Ko Ko	2000.00